A Small Transaction Path

A three-page fixture with a figure and a table

1 Overview

Clients send their commits to a proxy. The proxy asks the resolvers whether the transaction conflicts with one committed after it started, then pushes the accepted writes to the log servers. Storage servers pull the log and apply it.

2 Architecture

The figure below shows the five roles and the path a commit takes between them. Each arrow is a network message; the label names what the message asks for.

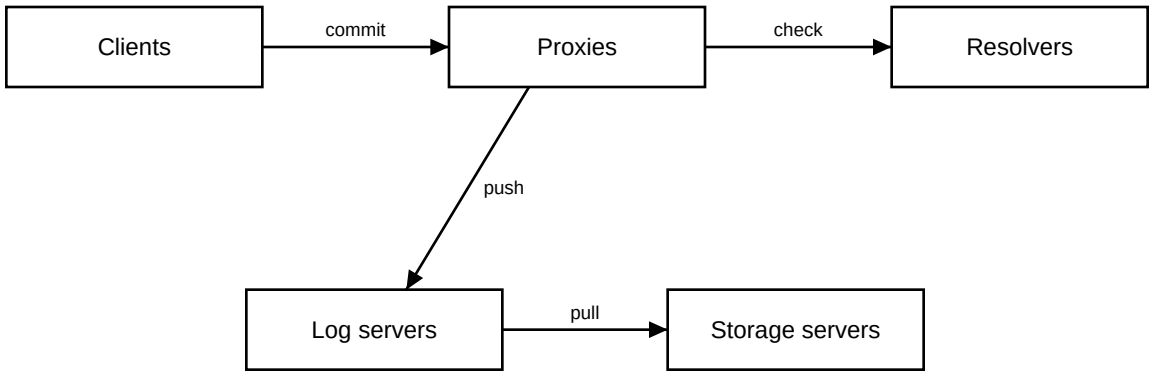

Figure 1: The transaction path from clients to storage.

A commit is acknowledged once every log server it was pushed to has made it durable. Storage servers lag slightly behind and catch up in the background.

3 Evaluation

We measured each operation on a three-machine cluster. Reads are served by the storage servers alone, so they are the fastest; commits pay for the resolver check and the log push, as Figure 1 shows.

Table 1: Latency by operation.

Operation	p50 (ms)	p99 (ms)
Get read version	0.4	1.2
Read	0.3	0.9
Commit	2.1	6.8
Range read (100 keys)	1.1	3.5